



A Green Energy Company

Business Plan

Nanodify, Inc.
1430 Tully Road, Suite 308
San Jose, CA 95122
www.nanodify.com

Contact:

info@nanodify.com
(408) 623-5211

The information in this document is confidential and is to be read only by authorized parties. Please refer to the confidentiality agreement for further details. This business plan is not an offering for security.



Table of Contents

1. EXECUTIVE SUMMARY	3
About Us	3
Mission	3
The Company	3
Company Stage	3
Market Opportunity	4
Product & Technology	4
Nano-Nickel Catalyst	4
Titanium Photocatalyst	5
Fuel Cell Module for Electric Vehicle	6
Sales & Marketing	6
Competitive Advantage	6
Financial Plan	7
Go-to-Market Strategy	7
ROI and Exit Strategy	7
Executive Management Team	7
Other Key Nanodify Company Representative	8
Advisor Boards	8
Keys Business and Financial Strategy	8
BUSINESS PLAN	9
2. BUSINESS PROPOSITION	9
Technology Overview	9
3. MARKET RESEARCH AND ANALYSIS	9
Market Size	10
Hydrogen Production	10
Fuel Cell	10
Target Hydrogen Industries	10



Government Support in Hydrogen Commercialization of Fuel Cell Vehicles and Hydrogen Stations10

4. PRODUCT OFFERING..... 11

Custom-tailored Nano Catalyst Materials for Hydrogen, Biomass, and Commodity Chemicals Industries11

Fuel Cell Module for Electric Vehicle.....11

5. BUSINESS MODEL 15

6. COMPETITIVE ANALYSIS 15

Hydrogen production by steam reforming pains15

Hydrogen production by electrolysis pains15

Fuel cell pains.....16

Nanodify Competitive Advantage16

Why Our Solution Is Better?16

7. SALES AND MARKETING STRATEGY..... 17

Potential Customers17

Competitors.....17

8. OPERATIONS 17

9. MANAGEMENT TEAM 18

Advisory Board20

10. RISKS AND ASSUMPTIONS 20

11. FINANCIAL MODEL AND STRATEGY 20

12. INVESTMENT STRATEGY..... 22

13. WHY INVESTMENT IN NANODIFY?..... 22

Why It Is Important?23

We have connected products in three markets23

14. Future Roadmap.....23



Future Hydrogen Economy	23
We enable the transition to renewable energy resources	25



Confidentiality Agreement

The undersigned reader acknowledges that the information provided in this business plan is confidential; therefore, the reader agrees not to disclose it without the written permission of an authorized agent of Nanodify, Inc.

It is acknowledged by the reader that information furnished in this business plan is in all respects confidential in nature, other than information which is in the public domain through other means and that any disclosure or use of same by reader, and may cause serious harm or damage to aforementioned parties.

This business plan is not to be copied or reproduced by any means without the sole written consent of an authorized agent of Nanodify, Inc.

Upon request, this document is to be immediately returned.

Signature

Date

Name (typed or printed)



1. Executive Summary

About Us

Nanodify® provides the breakthrough synthesis of nano catalysts for hydrogen production, fuel cell, commodity chemicals, and biofuel from renewable feedstocks. Nanodify's key focus is in fuel cell technology for electric vehicle and consumer power generation using in-house high performance, high surface area, and high electro-catalytic catalysts. Nanodify believes to have the technology breakthrough in photocatalysts and nano nickel catalysts for use in hydrogen production and fuel cell. Nanodify has patent pending Titanium Photocatalyst filed by Morgan, Lewis & Bockius LLP.

Mission

Nanodify® mission is to be a leading supplier of nano catalysts for fuel cell, steam reforming, and biomass reforming companies. Nanodify's catalysts and fuel cell technologies can be a vehicle for reducing the dependency on fossil fuel. Nanodify aims to become a leader in the clean and renewable energy, and more specifically in fuel cell for electric vehicle and consumer power generation.

The Company

Nanodify® is a Delaware Corporation, established in May, 2010 through the combination of CPNano, Inc., which founded in 2005. Nanodify is an advanced nano materials manufacture company that specializes in custom tailored nano materials. Since 2005 and through continuous research and development, Nanodify has revolutionized Nano-Nickel catalyst and Titanium Photocatalyst for hydrogen production and fuel cell. Nanodify's fully equipped research and development laboratory headquarter is located in the Silicon Valley, California.

Company Stage

Nanodify's products are ready to market: Nano-Nickel catalyst and Titanium Photocatalyst. Nanodify's Titanium Photocatalyst is low-cost, high performance material for stream reforming, and biomass reforming into commodity chemicals and biofuel. Nanodify's Nano-Nickel catalyst is a high performance, high surface area, and low-cost material for hydrogen production and fuel cell.



Market Opportunity

Nanodify®'s initial market targets companies that manufacture commodity chemicals, biofuel, and hydrogen production. Nanodify® primary market is fuel cells, electric vehicles, and consumer power generation. Nanodify's Nano-Nickel catalyst and Titanium Photocatalyst are a multi-billion-dollar industry. According to Lux Research (2009), the market size for nanomaterials and nano-enabled products could reach \$3 trillion by 2015. The current market size for this sector is only 2% of the \$3 trillion dollars.

Nickel-based catalysts are well-known and commercialized products under the trade names Raney nickel™, Sponge Metal™, Skeleton nickel, Actimet™, etc. Nanodify's high surface area Nano Nickel catalyst can significantly improve the efficiency of steam reforming, fuel cell, electrolysis, hydrogenation, ammonolysis, reductive alkylation, and dehydrogenation of organic compounds for the fibers, pharmaceuticals, plastics, perfumes, soaps, color couplers and the petroleum industries.

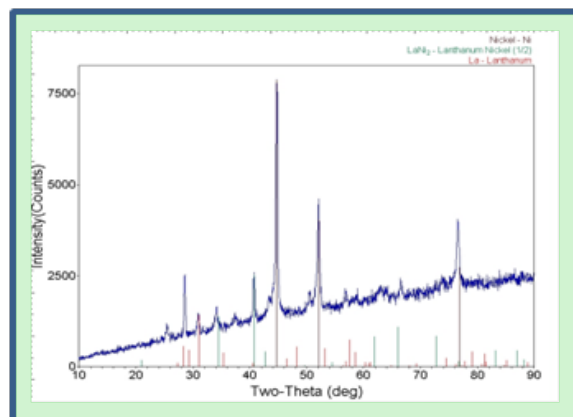
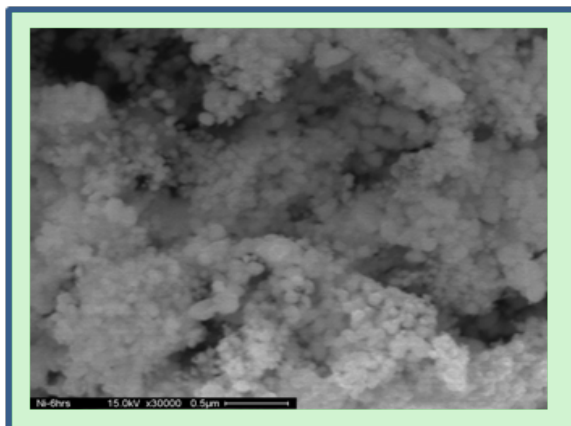
Product & Technology

Nanodify® provides three different products/technologies: 1) Titanium Photocatalyst for commodity chemicals and biofuel from renewable feedstock, 2) Nano-Nickel catalyst for fuel cells in electric vehicles and for consumer power generation, hydrogen production, and 3) Fuel cell module for electric vehicles and consumer power generation.

Nano-Nickel Catalyst

Nanodify's Nano-Nickel Photocatalyst is a real value driver in the development of fuel cell and in hydrogen production processes. Our proprietary process produces custom-tailored nano-nickel catalysts for the specific needs of our customers' catalytic selectivity application. Nanodify produces particle size in the range of 20-50 nanometers, and a surface area has the range of 40–60m²/gm. Nanodify's nano-nickel is an excellent electrical conductance material in support of hydrogen production and fuel cell.





Specification:

<i>Chemical Formula</i>	<i>Ni\Ni_xLa\NiO_x</i>
<i>Crystal Phase</i>	<i>Crystal \FCC</i>
<i>Appearance</i>	<i>Black</i>
<i>Particle Size Range</i>	<i>below 50nm</i>
<i>Specific Surface Area (BET)</i>	<i>40-60m²/gm</i>
<i>Bulk Density</i>	<i>0.15g/cm³</i>
<i>Ni Density</i>	<i>8.89g/cm³</i>

Titanium Photocatalyst

Nanodify's core-shell photocatalyst materials are custom tailored heterogeneous catalyst for biomass reforming and sugar derived chemicals. Our sugar technology provides a cost-effective catalyst for used in the production of hydrogen, commodity chemicals, and biofuel as a renewable alternative solution.





Specification:

<i>Chemical Formula</i>	<i>MTiO_x</i>
<i>Crystal Phase</i>	<i>Anatase/Rutile</i>
<i>Appearance</i>	<i>Colored</i>
<i>Particle Size Range</i>	
<i>Specific Surface Area (BET)</i>	
<i>Bulk Density</i>	

Fuel Cell Module for Electric Vehicle

Nanodify® provides the design of a fuel cell module for electric vehicles and consumer power generation through Nanodify's Nano-Nickel catalyst material. Nanodify's Nano Nickel catalyst is a low-cost catalyst with high surface area, high performance, and high electro-catalytic reactivity for hydrogen production and fuel cells.

Sales & Marketing

Nanodify® is looking to generate revenue through sales, partnerships, and technology licensing of Nano-Nickel catalyst, Titanium Photocatalyst, and custom-tailored nano catalysts. Nanodify is also providing fuel cell modules for electric vehicles and looking for a joint venture or co-developing partner for the next generation of fuel cells and consumer power generation.

Competitive Advantage

Nanodify's key competitive advantage is our ability to develop and produce low-cost, high-quality metallic nanopowder catalysts across a wide range. Catalysts are a key driver in steam reforming and biomass reforming processes for hydrogen production. It is estimated that hydrogen production and fuel cells are on the rise. Nanodify envisions an exponential growth potential in hydrogen production and fuel cells for electric vehicles and power generation by 2014.

Nanodify's catalyst materials can be used in steam reforming, which converts natural resources such as coal, natural gas, crude oil, and biomass into hydrogen, with the majority of the hydrogen used in petroleum refinery processes, fuel cells, and fertilizer production to meet agricultural demand and feed our growing global population. Nanodify's catalyst materials provide real value to hydrogen production, fuel cells, and reforming industries. It is expected that catalyst materials play a significant role in increasing the efficiency of power generation.

In laboratory testing, Nanodify's nano-nickel-based catalyst shows high reactivity and is well-suited to supporting steam reforming, biomass reforming, hydrogen production, and fuel cells. Nanodify's competitive, cutting-edge nano-nickel-based catalyst sales potential is expected to be a multi-million-dollar industry. It is important to note that Nanodify's catalysts are low-cost and highly scalable, supporting mass commercialization.



Financial Plan

Nanodify® is currently seeking funding of \$5 million dollars to enhance Nanodify's Photocatalyst technologies and to develop a fuel cell module for electric vehicles and consumer power generation. Nanodify is also aggressively pursuing government loan guarantees and government grants from the U.S. Department of Energy (DOE). In that case, Nanodify may require less than the target funding.

Go-to-Market Strategy

Nanodify® go-to-market strategies will be through sales, partnership, and technology licensing. Our primary focus will be as follows:

- Nano-Nickel catalyst, Titanium Photocatalyst, and custom-tailored nano catalysts
- Technology development of hydrogen production through steam and biomass reforming
- Fuel cell module for electric vehicle

ROI and Exit Strategy

Nanodify® believes there will be an excellent Return on Investment (ROI) from our Nano-Nickel catalyst, Titanium Photocatalyst, and fuel cell technologies. We are targeting ROI either through an acquisition or IPO within 6 years.

Executive Management Team

Nanodify® currently has 8 core team members who possess impressive track records in nanomaterials science, biochemistry research and development, and engineering. The team is committed to making Nanodify a successful American company, and, most importantly, we are a cohesive and results-oriented team. We are actively recruiting top-notch engineers and scientists to work on fuel cell projects and to continue the development and improvement of Nanodify's core-shell photocatalyst technologies. Please see Management Team biographies in Section 9.

- Dr. Jim Pham, Ph.D, CEO
- Vincent Pham, CTO & Founder, Scientist
- Dr. Ang-Ling Chu, Ph.D., Vice President of Operations & Scientist
- Van-Hanh Nguyen, MBA, J.D., CFO
- John Zajac, Vice President of Engineering
- Tom Huynh, Vice President of Production
- Nhut Tran, Director of Business Development
- Dr. Shao-Sheng Dong, Ph.D., Polymer Scientist



Other Key Nanodify Company Representatives

- Patent Counsel: Morgan, Lewis & Bockius LLP
- Corporate Attorney: Morgan, Lewis & Bockius LLP
- Analytical Chemistry Service: Dr. J. P. Hsu, Ph.D., Smart Chemistry Corporation

Advisor Boards

Nanodify® advisor boards are well-known academic scholars and industry leaders from various disciplines.

- Dr. Joseph Adwere-Boamah, Ph.D., Academic Scholar, UC Berkeley
- Dr. Jason Huang, Ph.D., Mechanical Engineering, Queen Mary, University of London
- Dr. Nobuyuki Kambe, Ph.D. from MIT; co-Founder of Nanogram Devices, Neophontics, and Nanogram Corporation.
- Dr. Tien D. Pham, Ph.D., University of Toronto in EE; co-Founder of Radilan
- Dr. Edward Shenk, Ed.D., Academic Scholar, University of Oregon
- Dr. George Stamos, Ed.D., Academic Scholar, Northern Arizona University
- Mr. Daniel Barr, MBA, Director of Transportation Systems for the State of Ohio
- Mr. George Reis, Engineering Executive, BSEE, Iowa State University
- Mr. Hung Le, Technology Veteran and Executive; MSEE and BSEE from MIT with specialization in device physics

Keys Business and Financial Strategy

1. Technology development and manufacturing custom-tailored heterogeneous catalysts for hydrogen production, fuel cells, biomass reforming and sugar-derived chemicals
2. Technology development for fuel cells in electric vehicles
3. Technology development for sugar-derived chemicals
4. Technology development for biomass reforming
5. Intellectual Property (IP) development
6. Revenue generation through sales, partnership, and technology licensing



BUSINESS PLAN

2. Business Proposition

Technology Overview

Nanodify® core technology is in the synthesis of nano catalyst. Nanodify believe to have the right chemistry, nanometer size, high surface area, and high catalytic reactivity for hydrogen production and fuel cell.



Nanotechnology and nano catalysts are drivers of steam reforming, biomass reforming, hydrogen production, and fuel cells. Nanodify's business is our ability to develop/produce low-cost and a wide range of high-quality metallic nano powder catalysts.

Nanodify's nano nickel-based catalyst production is low-cost, versatile, and completely scalable. Nanodify's competitive advantage lies in our ability to develop high-surface-area, superior nanoparticle-size synthesis, high performance, and low cost. Nanodify's niche area of expertise is our technology to dope well-known catalyst promoters such as the elements in the alkali metals, alkaline earth metals, and lanthanides to optimize the performance of catalytic materials and to fine-tune formulations to the specific needs of our customers' catalytic selectivity.

3. Market Research and Analysis

According to Lux Research (2009), nanotechnology will have a significant impact on the energy and clean-tech sector. The U.S. Congress (2009) has passed the stimulus bill worth more than \$90 billion dollars targeting clean energy, the largest investment in the energy sector. The size of the market for fuel cells, hydrogen production, and catalyst materials is a multi-billion-dollar industry. Nanodify will focus primarily on fuel cells for electric vehicles and hydrogen production through Nanodify's Nano-Nickel catalyst and Titanium Photocatalyst. The market size for nanomaterials and nano-enabled products could reach \$3 trillion by 2015. The current market size for this sector is only 2% of \$3 trillion.



Market Size

Hydrogen Production

- For Industrial Uses:
 - Worldwide in 2006 – 48 billion KG
 - US in 2008 – 20 billion KG
 - Typical price is \$2 to \$12 per KG
- For Transportation
 - The United States has an ambitious goal of displacing 35 billion gallons of annual petroleum use with renewable and alternative fuels by 2017
 - Corporate Average Fuel Economy (CAFE) regulations forcing the increase of alternative fuel usage by Car Manufacturers
 - The California Fuel Cell Partnership has released a detailed plan for a coordinated rollout of hydrogen fueling stations

Fuel Cell

- 600 million vehicles
- Growing by 30% in the next 10 years
- The need for electricity is projected to grow by 50% in the next 20 years

Target Hydrogen Industries

Hydrogen production is vital to the growth and stability of our global energy needs. The consumption of hydrogen in the USA alone was \$135 billion in 2005 and increased at a rate of over 10% per year. The production of ammonia (NH₃) for agricultural uses, such as fertilizer, accounts for over 50% of global hydrogen production. Nearly 50% is used in petroleum refineries process, chemical synthesis, and fuel cell applications. The production of hydrogen by the steam reforming process accounts for approximately 96%: 48% from natural gas resources, 30% from crude oil, and 18% from coal. The remaining 4% is through electrolysis.

Government Support in Hydrogen Commercialization of Fuel Cell Vehicles and Hydrogen Stations

There is a lot of support from the U.S. government in clean, alternative, and renewable resources. As a result, the Department of Energy (DOE) and the Department of Transportation (DOT) are aggressively investing in the development of hydrogen technology and the commercialization of fuel-cell-powered ground support equipment for commercial and government-operated airports. Under the Bush-Administration, the Hydrogen Fuel Initiative (HFI) was passed, granting \$1.2 billion in funding. Under the Obama Administration, there is overwhelming support for clean and alternative energy resources to reduce greenhouse gas emissions and stimulate job creation. Recently posted on June 8, 2009, with a closing date of August 24, 2009, A “Clean Coal Power Initiative” was granted to support commercialization of coal-to-gas for energy needs that will



reduce greenhouse gas emissions from the current standard coal-burning plant method, which supplies about 51% of our country's electricity.

4. Product Offering

Nanodify® currently has products ready to market. We are looking for sales, partnership, and technology licensing for our products.

Custom-tailored Nano Catalyst Materials for Hydrogen, Biomass, and Commodity Chemicals Industries

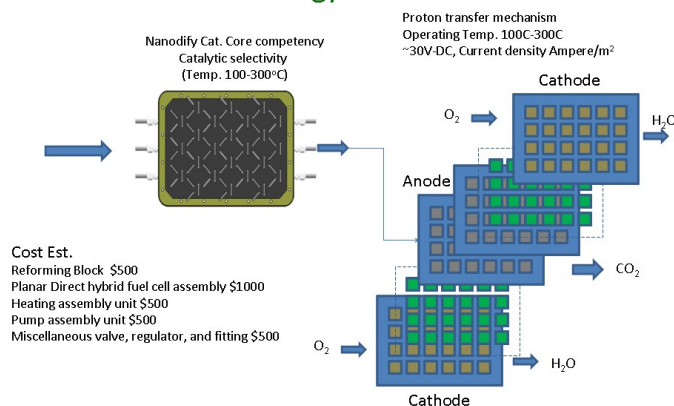
- Nano-Nickel
- Nano-Nickel-based (Customer-tailored product)
- Nano-Nickel-based iron oxide
- Nano-Nickel-based potassium
- Nano-Nickel-based molybdenum oxide
- Nano-Nickel-based lanthanum oxide
- Nano-Nickel-based zirconium oxide
- Nano-Nickel-based cerium oxide
- Nano-Nickel-based multi-metal oxides
- Nano-Nickel-based precious metal group (such as platinum, ruthenium, or palladium)
- Nano-Nickel catalyst for fuel cell
- Titanium Photocatalyst for biomass reforming

Fuel Cell Module for Electric Vehicle

Nanodify® has designed a fuel cell module for electric vehicles using our high-performance, highly catalytic materials. Nanodify's Nano-Nickel catalyst is a low-cost/real-value ROI catalyst than high-cost platinum, gold, and other much more expensive catalysts. We are expected to deliver a real fuel cell module by 2013. Nanodify's fuel cell technology and steam reforming process are as follows:

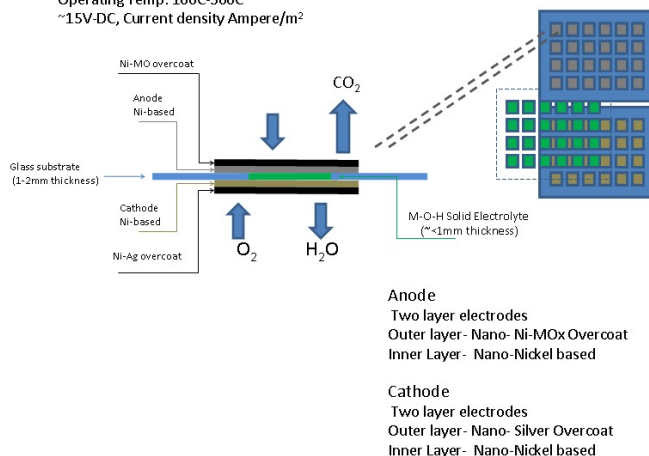


Fuel Cell Technology



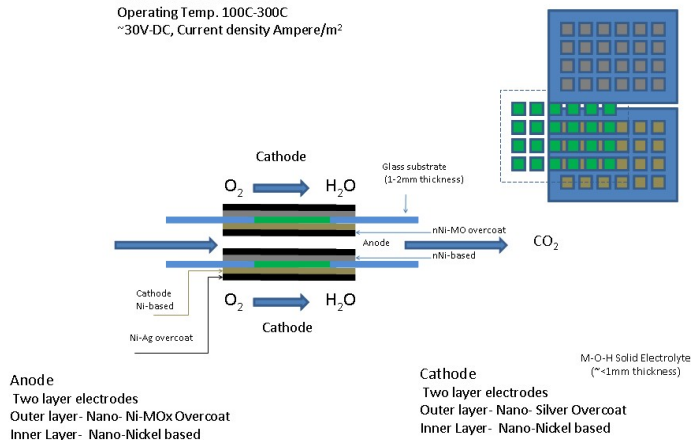
Cross Section View Planar Fuel Cell

Proton transfer mechanism
Operating Temp. 100C-300C
~15V-DC, Current density Ampere/m²

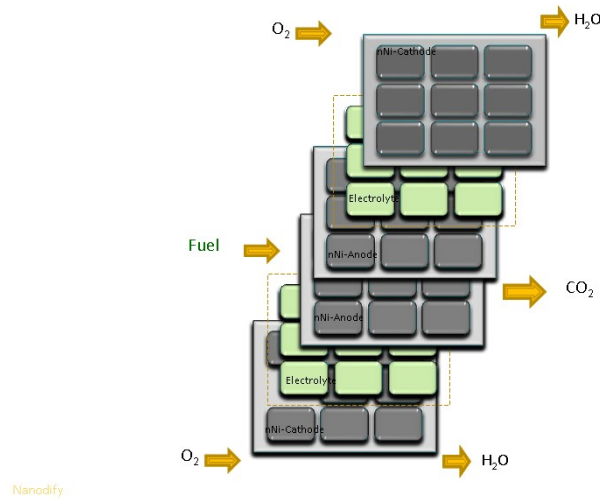


Cross Section View Planar Fuel Cell

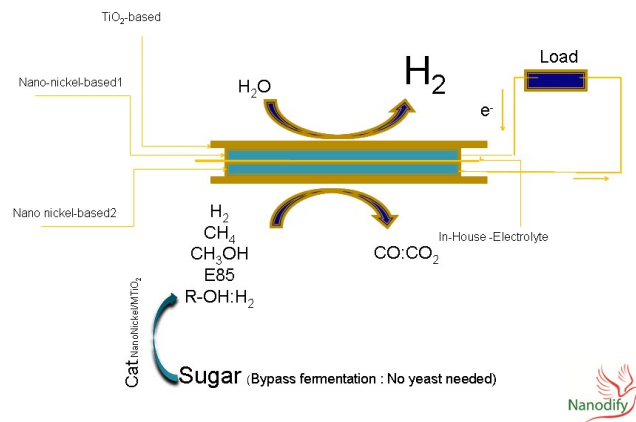
Proton transfer mechanism
Operating Temp. 100C-300C
~30V-DC, Current density Ampere/m²



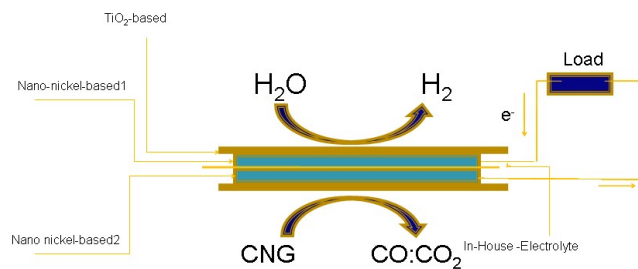
Alkali/Molten Carbonate Fuel Cell



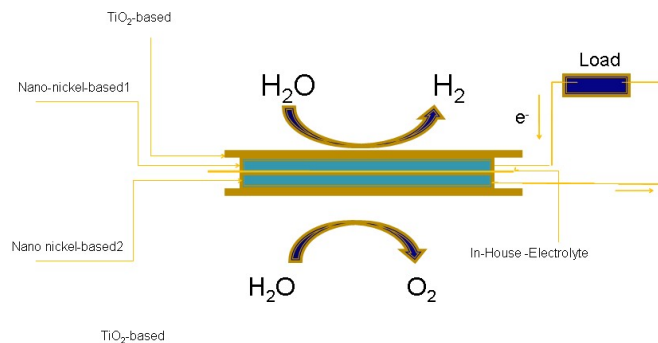
Nanodify Fuel Cell Electrolyzer



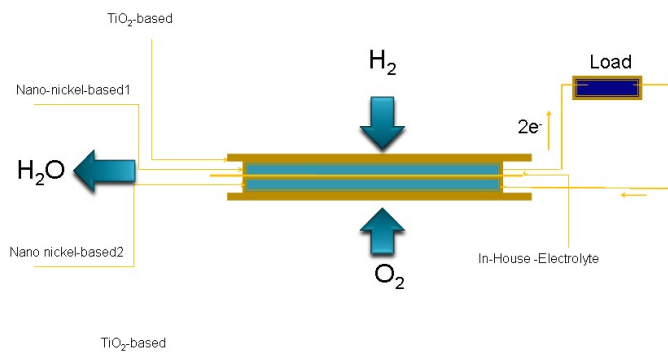
Nanodify Hydrocarbon Fuel Cell Electrolyzer



Nanodify Water Fuel Cell Electrolyzer

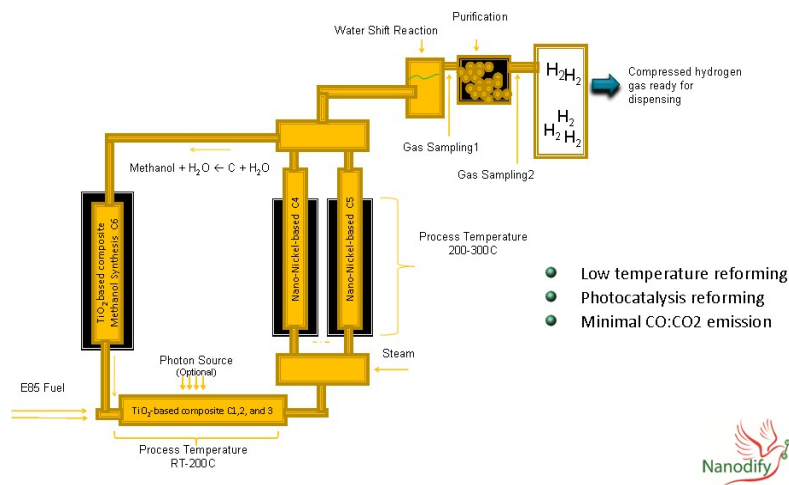


Nanodify Hybrid Fuel Cell Electrolyzer



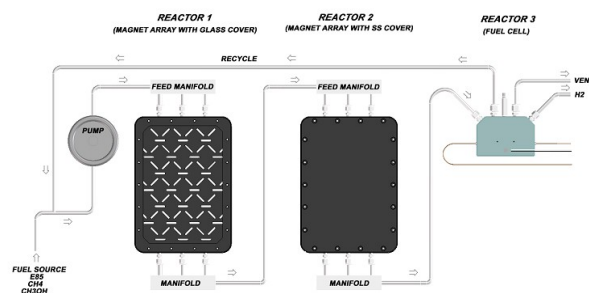
Nanodify Reforming Process

A Low Temperature Hydrogen Production





Nanodify Reforming Process Natural Gas $\rightarrow$ H₂dirty $\rightarrow$ H₂purified



5. Business Model

Nanodify's business model is based on our advanced research and development of nano catalyst materials for hydrogen production and fuel cells. Nanodify focuses on the following areas:

- Engineering and manufacturing custom-tailored nano catalysts for hydrogen production, fuel cells, steam reforming, biomass reforming, and sugar-derived chemicals
- Fuel cell development for electric vehicles
- Joint venture in co-developing the next-generation fuel cell for electric vehicles and consumer power generation
- Intellectual Property (IP) development
- Revenue generation through sales, partnership, and technology licensing

6. Competitive Analysis

Hydrogen production by steam reforming natural gas

- Steam reforming process suffers from low efficiency (50-60%) due to catalytic poisoning.
- The operation cost is high.
- Greenhouse gas emissions from the process are high.

Hydrogen production by electrolysis pains

- Expensive precious metals are used as electrodes.
- Electrolysis is a costly approach to produce hydrogen.
- Electrolysis commercialization is limited.



- Efficiency is low.

Fuel cell pains

- Expensive precious metals are used as electrodes.
- Fuel cell stationery and portable devices are too costly.
- Fuel cell commercialization is limited.
- Ultra-pure gases or liquid fuels are not readily available to support fuel cells.
- There is no infrastructure to support fuel cell commercialization.

Nanodify Competitive Advantage

Nanodify's key competitive advantage is our ability to develop/produce low-cost catalyst materials and a wide range of high-quality metallic nano powder catalysts. Catalyst is a key driver in steam reforming and biomass reforming processes for hydrogen production. It is estimated that hydrogen production and fuel cells are on the rise. Nanodify envisions an exponential growth potential in hydrogen production and fuel cells for electric vehicles and power generation by 2014.

Nanodify's catalyst materials can be used in steam reforming that convert natural resources such as coal, natural gas, crude oil, and biomass to hydrogen for majority usage in petroleum refinery processes, fuel cells, and fertilizer production for agricultural demand to feed our growing global population. Nanodify's catalyst materials provide real value to hydrogen production, fuel cells, and reforming industries. It is expected that catalyst materials play a significant role in increasing the efficiency of power generation.

In laboratory testing, Nanodify's nano-nickel-based catalyst shows high reactivity and is well-suited to supporting steam reforming, biomass reforming, hydrogen production, and fuel cells. Nanodify's competitive, cutting-edge nano-nickel-based catalyst sales potential is expected to be a multi-million-dollar industry. It is important to note that Nanodify's catalysts are low-cost and highly scalable, supporting mass commercialization.

Why Is Our Solution Better?

Nanodify's breakthrough nanocatalyst technology delivers real value and ROI for both hydrogen production and fuel cell applications.

- Much lower cost than most existing solutions such as platinum, gold, etc.
- A catalyst-based system that is completely scalable
- Low-cost production
- Easier to install in a device that has a small form factor
- Requires less energy to operate



7. Sales and Marketing Strategy

Nanodify® believes that we can achieve market penetration as early as 2012. We intend to operate on sales, partnership, and technology licensing. Our hope is in collaborating with the government, partnership venture to co-develop in next generation of fuel cell for electric vehicle and consumer power generation.

Potential Customers

The U.S. Department of Energy (DOE), Hydrogenics, Intelligent Energy, Horizon Fuel Cell, Nuvera, Relion Inc., UltraCell Power, UTC Power, Trulite, PlugPower, Ballard Power System, IdaTech, GreatPointEnergy, Renetech, Air Product, Air Liquide, Praxair, Dupont Fuel Cell, Global Hydrogen, ITM Power, H2gen, Panasonic, Toyota, GM, etc.

Competitors

Unlike our competitors, Nanodify's catalyst technology is low-cost versatility, and completely scalable. Our competitors include: QSI Nano, BASF, Johnson Mathey, Grace, CRI Catalyst, Matrosteck, Sued Chemie Catalyst, Umicore, Inco, Haldor Topsoe, Nextech Materials, Headwaters Technology, Linde Ag, Cabot Corporation, Integran Technology, and N.E. Chemcat.

8. Operations

Nanodify's research and development headquarter is located at Tully Business Center, 1340 Tully Road, Suite 308, San Jose, California, 95122. The Tully Business Center is well-suited for conducting nanomaterials research and development, hydrogen production, and fuel cell research. The Tully Business Center was designed as a high-tech facility with a modular research laboratory, chemical permits, on-site safety, facility support, and security. Nanodify's leases approximately 1728 sq. ft. of space at this location. Our research laboratory is fully equipped with Agilent/HP GC-MS, RKI Eagle gas detector, Thermoline 30400 furnaces (x2), two Thermoline 48000 furnaces (x2), Lindberg Blue M furnaces (x2), Riko Kiden 700S high-temperature furnace, etc.



1340 Tully Road, Suite 308, San Jose, CA



9. Management Team

Nanodify® currently has 8 core team members who possess impressive track records in nano materials science, biochemistry research and development, and engineering. The team is committed in making Nanodify a successful American's company and most importantly, we are a cohesive and result-oriented team. We are actively recruiting top-notch engineers and scientists to work on fuel cell projects and to continue the development and improvement of Nanodify's core-shell photocatalyst technologies.

Jim Pham, CEO

Jim Pham has over 18 years of experience in research and development across computer hardware/software design, enterprise application software design, and educational research. Dr. Pham also has over 12 years of experience in academia. Prior to Nanodify, Dr. Pham was a co-founder of Radilan and an engineer at Oracle, Lockheed, Elonex, Diamond Computer, and Hyundai. Dr. Pham's areas of expertise include hardware/software design, engineering, and quantitative/qualitative research. Most notable accomplishment: Dr. Pham was awarded the NASA Ames Honor Award, the agency's most prestigious award, for his work on the Beacon Processing System for Pharmsat (a NASA nano satellite, June 2009). Dr. Pham also spoke at several conferences and, most recently, presented a case study on podcasting at TechEd 2010. Dr. Pham holds a bachelor's degree in Computer/Electrical Engineering from San Jose State University, a master's degree in Software Engineering from National University, and a doctorate in Higher Education from Shirley M. Hufstедler School of Education.

Vincent Pham, CTO & Founder

Vincent Pham has had over 10 years of advanced nanomaterials research and development. Mr. Pham is a gifted self-achiever in surface chemistry, nano dispersion, and nanomaterials formation. Prior to Nanodify, Mr. Pham was an Engineer for Sipix Imaging, Nanogram, and Trillion Science. Mr. Pham holds a bachelor's degree in Chemistry from San Jose State University. Mr. Pham's strength lies in colloidal chemistry and flame hydrolysis processes for developing advanced nanomaterials. Mr. Pham has developed a dispersion process using metal and metal-oxide nanoparticles for thin film coating and printed-electronics applications. Mr. Pham holds 2 provisional patents.

Ang-Ling Chu, Vice President of Operations & Scientist

Ang-Ling Chu has over 20 years of experience in electronic materials and polymer chemistry for semiconductors and electronic applications. Prior to Nanodify, Dr. Chu was a Vice President of Polymer Research at Isis Biopolymer and an Applications Manager for Nanocomposite at NanoGram. Dr. Chu holds a Ph.D. from Rensselaer Polytechnic Institute (RPI) and a B.S. from



National Tsing Hua University. Dr. Chu has published several technical papers and holds 4 U.S. patents.

Van-Hanh Nguyen, CFO

Van-Hanh has over 28 years of experience in accounting and 30+ years of service to the community in Oakland. Hanh owns an accounting firm, Van-Hanh Nguyen, CPA. Hanh holds a J.D. degree from Concord Law School and an MBA from San Francisco State University.

John Zajac, Vice President of Engineering

John Zajac is an inventor-entrepreneur with over 35 years of experience in advanced technology employing “Out of the Box” creativity. Mr. Zajac is a co-Founder of ZajacMotors and has held several executive positions: VP of R&D, ZOOM; Product Manager, Mattson Technology; Founder & Executive VP, Results Corps; VP & General Manager, E. T. Systems. Mr. Zajac has proven success in implementing productivity-enhancement systems and an entrepreneurial approach to identifying and developing advanced technology. Mr. Zajac holds a BSEE from the University of Kentucky and has 40 U.S. patents and 10 pending.

Tom Huynh, Vice President of Production

Tom Huynh has over 15 years of EHS and facilities design and operations experience in high-throughput materials research, semiconductor, and electronic facilities. Prior to Nanodify, Mr. Huynh was a Director of EHS at Symyx Technologies. Prior to Symyx, Mr. Huynh was EHS Manager at Applied Materials, Sr. Facilities and Safety Engineer at CPI (a former division of Varian Associates), and Senior Consultant at CBA (a facilities and safety consulting company for healthcare and university medical centers). Mr. Huynh holds a B.S. in Chemical Engineering and is a registered PE in the State of CA. Written and published 7 technical papers and spoke at 4 major professional conferences.

Nhut Tran, Director of Business Development

Nhut Tran has over 20 years of experience as a Database Administrator (DBA). He managed critical databases (Oracle, DB2) for large companies such as Charles Schwab, Gap & Siebel. He currently works as a Lead & Principal DBA at Oracle. In his role, he works & review code with engineers to ensure they are optimized for performance. Prior to his time as a DBA, Mr. Tran held various engineering management and project leader roles at a software Retail company, PRJ. Mr. Tran is a graduate of Adelaide University in Australia with a degree in Computer Science and Electrical & Electronic Engineering.



Shao-Sheng Dong, Polymer Scientist

Shao-Sheng Dong is a research scientist from Symyx Technologies, who has extensive knowledge in biochemistry research. Dr. Dong holds a Ph.D. in Chemistry from Case Western Reserve University.

Vice President Sales and Marketing

To be determined

10. Risks and Assumptions

Nanodify® has been a privately funded company since 2005 and is subject to potential risks arising from a variety of factors common to high-tech companies. The risk assessment is from low to medium. The risks are technical, market, competitive, and financial.

In terms of technical risk, Nanodify has products ready to market — Nano-Nickel catalyst and Titanium Photocatalyst — for hydrogen production and fuel cells. However, there are challenges in deploying hydrogen and fuel cell technologies.

In terms of market risk, the current market size for this sector is only 2% of the \$3 trillion dollars. There is tremendous potential for ROI.

In terms of the competitive risk, there are other players. However, Nanodify believes it has a technological breakthrough in photocatalysts and nanonickel catalysts for use in hydrogen production, fuel cells, and mass commercialization.

In terms of financial risk, Nanodify has been a privately funded company for the past 5 years. Nanodify is hopeful about being funded either by U.S. government grants and loans or by venture capital. The U.S. Congress (2009) passed the stimulus bill worth more than \$90 billion, targeting clean energy.

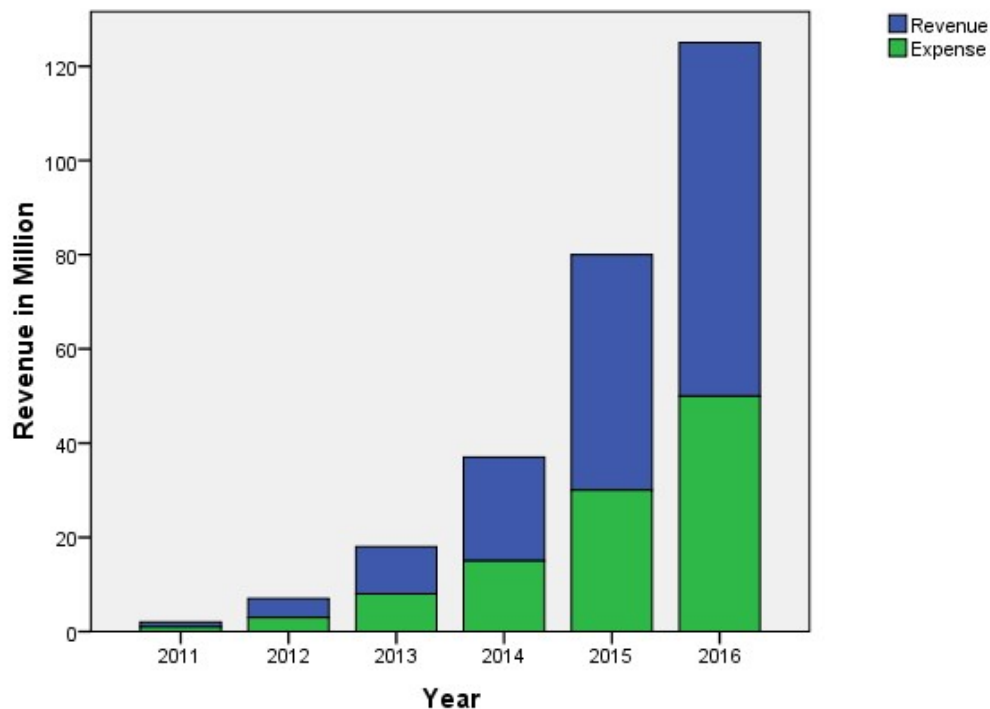
11. Financial Model and Strategy

Nanodify's financial model is through sales, partnerships, and technology licensing. In the green energy space, there is growing investment by the U.S. Department of Energy (DOE). With government grants/loans and sales, Nanodify would be a self-sustaining company by 2012. By 2014, Nanodify's financial projection will be profitable and have over \$50 million dollars surplus.



Nanodify Financial Projections - Four Years Plan				
	Year 1 - 2011	Year 2 - 2012	Year 3 - 2013	Year 4 - 2014
Expenses	\$2M	\$3M	\$8M	\$15M
o Salaries/Overhead	\$600K (7+ Emps)	\$1.5M (10+ Emps)	\$4.5M (25+ Emps)	\$8M (40+ Emps)
o Equipment	\$100K	\$300K	\$500K	\$1M
o Sales & Marketing	\$200K	\$600K	\$1M	\$2.5M
o Misc	\$50K	\$100K	\$500K	\$500K
o Operations	\$50K	\$200K	\$500K	\$1M
o Manufacturing	\$0	\$300K	\$1M	\$2M
Cash Inflow				
o Venture Financing	\$2M	\$3M	\$0	\$0
o Grants & Loans	\$0	\$2M	\$2M	\$2M
o Total (Sales, Partnership, Technology Licensing)	\$0	\$0	\$6M	\$20M
- Fuel Cell Module for electric vehicle			\$2M	\$15M
- Nano-nickel and custom-tailored nano catalysts			\$4M	\$10M
Surplus		\$1M	\$3M	\$7M

Revenue Projection





12. Investment Strategy

Nanodify® estimates that it will be able to self-sustain through sales, partnerships, and technology licensing by 2012. A Series-A investment of \$5 million dollars is sought from various investment sources. The initial funding will enable us to build company infrastructure, build a core team, conduct technology validation with experimental data, and develop improvements. Nanodify is also pursuing guaranteed loans and government grants from the U.S. Department of Energy (DOE). These funds primarily will be used for research and development of nano catalyst materials for hydrogen production and fuel cell modules for electric vehicles and power generation. Nanodify will focus primarily on the following investment strategy:

- Technology development and manufacturing custom-tailored heterogeneous catalysts for hydrogen production, fuel cells, biomass reforming, and sugar-derived chemicals
- Technology development for fuel cells in electric vehicles and consumer power generation
- Technology development for sugar-derived chemicals
- Technology development for biomass as reforming
- Intellectual Property (IP) development

13. Why Investment in Nanodify?

Nanodify® provides an excellent ROI in the next big wave in the technology shift--nano-enabled products and clean and renewable energy sources. Nanodify understands, researches, and develops a real energy solution that can significantly help reduce dependence on fossil fuels. Nanodify offers investors the potential for a huge return on their investment in our breakthrough technology solutions. Most importantly, Nanodify's technologies offer real value: low-cost, high-performance, and scalable catalyst materials for fuel cells and hydrogen production. Nanodify's initial target markets are steam reforming, biomass reforming, and fuel cells for electric vehicles. According to Lux Research (2009), the huge market size for fuel cells, hydrogen, biomass, and nano-enabled products is in the multi-trillion-dollar range. Nanodify's key-benefits are as follows:

- Hydrogen/fuel cells are a more practical solution for electric vehicles than currently available.
- Nanodify's revolutionary low-cost catalyst materials are a cost-effective approach for hydrogen production, fuel cells, and electrolysis.
- Nanodify provides fuel cell modules for electric vehicles using all Nanodify's technologies and materials.
- Nanodify's technology can be used in power generation.
- Nanodify's Nano-nickel-based catalyst is low-cost compared to high-cost platinum and other precious metal catalysts.
- Nanodify's Nano-nickel-based catalyst delivers high conversion efficiency at low cost.
- Nanodify's Nano-nickel-based catalyst provides a practical solution for hydrogen production.



- Nanodify's Titanium Photocatalyst materials for biomass reforming to hydrogen gas.

Why Is It Important?

We have connected products in three markets

- Steam reforming and biomass reforming into hydrogen
- Fuel cell module for electric vehicle
- Power generation for homes and offices

14. Future Roadmap

Nanodify®'s revolutionary technologies can be a vehicle for meeting future energy needs in transportation and power generation.

Future Hydrogen Economy

- As a society, we need to move to a combustion-free transportation future
- Hydrogen is the most abundant, renewable, clean energy resource
- Hydrogen and fuel cells are a realistic solution for a single-fuel-type clean-powered vehicle
- Carbon footprint of automobile companies is being mandated, and they have no other solution at this time.
- Commercialization in hydrogen production by electrolysis will reduce the need for steam reforming processes attributed to greenhouse gas emissions.
- Electrolysis can be installed practically in any location to produce on-site and on-demand hydrogen fuel.
- The hydrogen economy infrastructure can replace existing infrastructure such as gasoline pump stations.
- Hydrogen infrastructure can support fuel cell commercialization in transportation vehicles and portable electronic devices.
- Commercialization in fuel cell transportation vehicles will significantly reduce greenhouse gas emissions.

We enable the transition to renewable energy resources

A green future!